

KLIMA Sense: Real-Time Mold Risk Prediction for German Residential Spaces

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Abstract—Exposure to indoor mold and dampness is a severe environmental health hazard which increases risk for respiratory diseases such as Asthma, bronchitis etc by 50%. Conventional indoor air quality systems are reactive and only address mold only after visible germination has occurred. This study investigates whether a low-cost, IoT-enabled mold prediction system can accurately forecast and proactively prevent indoor mold risk through the continuous monitoring of temperature and humidity. To answer this, a distributed edge computing architecture was developed using Nordic nRF 52 Series microcontrollers powered by Zephyr RTOS and CoAP OpenThread mesh network. The predictive core relies on an embedded implementation of the VTT mathematical mold growth model which dynamically calculates mold index accumulation over time without relying on continuous cloud processing. To validate the firmware and real-time scheduling, a custom time accelerated Simulation Engine was developed. The results show that the system successfully tracks aggressive mold accumulation and proactively triggers preventative warnings before the critical active growth threshold is reached. Furthermore, under severe time compressed and computational stress, the real-time software scheduling proved highly robust, successfully processing high-speed CoAP payloads and meeting strict 1000ms execution deadlines with 0% packet loss.

I. INTRODUCTION

People spend majority of their lives indoors which makes indoor air quality a critical factor for the human health. One of the most common and dangerous problems in indoor environments is the mold growth. Exposure to residential mold and dampness has severe health effects, particularly for the respiratory system. A major systematic review and meta-analysis revealed that living in damp and moldy homes increases the risk of

developing Asthma by 50% on average[1]. This risk can be higher depending how the mold is detected. Smelling a mold odor increases the asthma risk by 73%, while seeing the visible mold increases the risk by 29%[1]. This makes the environment dangerous for the vulnerable group. For example, infants living in moldy environments face a 99% increased risk of developing asthma, while children face a 30% increase and adults face a 19% increase [1].

As mold causes so many respiratory diseases, it creates a massive health and economic burden on society. In Europe alone, approximately 84 million people live in damp or moldy homes [1]. The Fraunhofer Institute reports that this increases the chance of developing asthma by 40%, along with a higher risk for allergies, bronchitis and skin conditions [2]. This health crisis leads to millions of missed school days for children, which lowers their grades and negatively impacts their future earning potential [2]. It also results in huge medical costs for hospital visits and medications. This is a global economic issue. In the United States, 35% of the homes have dampness and mold problems. This costs the society estimated to be \$22.4 billion every year [3]. The majority of this cost, about \$15.1 billion, is spent just on treating asthma caused by indoor dampness [3].

In Germany, the root cause for indoor mold growth is directly linked to the building structure and human behavior. To remove the indoor moisture proper ventilation is required but German residents face a unique barrier to do this correctly. A research shows that only 29% of the German residencies

have windows that can be easily opened for "Shock Ventilation" (opening the window fully for a short time) [4]. This is because German windows open inward and the window ledges are usually blocked by indoor plants or decorative items [4]. Due to this residents usually rely on "Kippstellung"—leaving the window tilted open for long periods. This titled window method is highly inefficient and consumes 20 times more heating energy than shock ventilation which fails to properly clear out indoor humidity [4]. When high indoor humidity is combined with this poor ventilation practice, it creates the perfect conditions for mold to grow [4].

Currently, the standard approach to deal with the indoor mold is reactive. People usually take action after the mold has already grown and became visible on the wall. By this time, the health risks are already present, and removing the mold is very expensive. A much better approach is to be proactive: predicting and stopping the mold before it ever becomes visible. To do this scientists use VTT Mold growth model which has been developed by the VTT Technical Research Centre of Finland. It calculates the risk by continuously looking at the temperature and relative humidity in a room over a period of time and accurately predicts the risk of mold growth on a scale of 0 to 6 [4]. Recent studies have proven that using ambient air sensors in homes to collect this data works perfectly with the VTT model [5].

Therefore, by combining the mathematical VTT model with modern and low-cost Internet of Things (IoT) sensors, it is possible to build a system that monitors the room environment and alerts the user to open the window before mold can grow, which leads us to the main research question of this project.

Research Question: *Can a low-cost, IoT-enabled mold growth prediction system accurately forecast indoor mold risk in German residential spaces through continuous real-time monitoring of temperature and relative humidity using an edge-computing IoT architecture?*

II. STATE OF THE ART

To fully review and understand the design of modern mold prevention system it is necessary to review the current research in several different

fields. This section discusses the transition from the tradition mold detection to proactive prediction, the mathematical models used to calculate the mold risk and the hardware and software technologies such as low cost sensors, edge computing, real-time operating systems and wireless mesh networks.

A. Traditional vs. Proactive Mold Detection

Mold is an ancient problem but the ways humans build homes have drastically changed our vulnerability towards it. In older times builders naturally incorporated mold fighting techniques into their homes. For instance, first century builders used mortar made of earth, chalk and hydrated lime. This naturally absorbed excess moisture and inhibited fungal growth [17]. However after the World War II, the construction industry shifted towards faster and cheaper materials. As a result Gypsum wallboard (drywall) became the standard building material. Because drywall consists of highly absorbent gypsum pressed between layers of paper, it creates perfect food and environment for mold to grow within 24 to 48 hours of water exposure [6]. This practice made indoor mold a much more common crisis.

The standard approach to indoor mold has historically been completely reactive. People usually wait until they see visible mold or smell a musty odor before taking action. Even professional mold detection has traditionally been a reactive process. To find hidden mold, inspectors often rely on thermal imaging cameras, which read infrared energy to locate cold and moist spots behind walls and under floors without tearing them apart [7]. Another traditional diagnostic tool is the Environmental Relative Moldiness Index (ERMI). It has been developed by the EPA and is a DNA based test that analyzes household dust to identify the history of mold species that have accumulated in a room over time [8]. While these tools are scientifically advanced, they are still fundamentally reactive and only confirm if it is a moisture problem or a mold infestation already exists [8]. After detection, homeowners are forced to perform expensive remediation, using chemical treatments or natural remedies like vinegar and grapefruit seed extract to clean the affected surfaces [9].

Moreover the traditional attempts to monitor and

control indoor moisture are flawed as they rely on static threshold. It is a common misconception that simply keeping a room's relative humidity below 60% guarantees that mold will not grow [10]. While the standard room hygrometer might show a safe humidity level in the center of the room, it fails to detect hidden "micro-climates" inside walls, under floors or in poorly ventilated corners where condensation builds up and creates perfect breeding grounds [10]. Another dangerous misconception in traditional mold management is the belief that simply drying out a room kills the mold but in reality, when mold loses its water supply, it does not die; it merely goes dormant [10]. These inactive spores act like tiny seeds, waiting for a next humid day or a minor leak to instantly resume their growth [10]. To truly solve this problem, modern scientific research has shifted from reactive cleanup and static thresholds, moving toward a fully proactive approach. The proactive strategy aims to stop the mold before it ever has a chance to grow by understanding the "mold growth triangle" of warmth, moisture and food. Instead of waiting to detect hidden water damage or measuring accumulated dust, the proactive system continuously monitors the dynamic environmental factors such as temperature, relative humidity and airflow in real-time. By continuously evaluating these factors, a proactive system can predict the exact moment when a room's micro-climate becomes dangerous, allowing residents to take simple actions, like turning on an exhaust fan or opening a window [10].

B. Mathematical Modeling for Mold Prediction (The VTT Model)

For a proactive system, scientists rely on established mathematical models that understand how mold biology works. The most widely recognized and validated model for building physics is the VTT Mold Growth model. This model has been developed by VTT Technical Research Centre of Finland. This model proves that mold growth is a dynamic process. Mold growth does not happen at a single temperature, instead it depends on temperature, relative humidity and time of exposure. The VTT model calculates the development of mold and expresses it as a "Mold Index" (MI) on a scale ranging from 0 to 6. An index of 0 means there

is absolutely no mold growth, an index of 1 means mold is starting to grow at a microscopic level and an index of 3 means mold is finally visible to the naked eye. The mathematical equation uses temperature and relative humidity data to continuously update this index. If the environment becomes dry, the model also accounts for the "decline" or dying off of the mold risk. Researchers have proven that this complex mathematical model can be successfully applied to real-world residential environments by feeding it continuous data from indoor air quality sensors.

C. IoT and Smart Home Sensors for Air Quality

To feed accurate data to the VTT model, highly reliable environmental sensors are required. In past, industrial air quality sensors were too expensive for standard homes. But thanks to the recent advancements in Internet of Things (IoT). It has led to development of low-cost miniaturized sensors that are highly capable of indoor air quality monitoring [16].

D. Edge Computing and Real-Time Operating Systems (RTOS)

Traditionally, smart home devices collected data and sent it to the central cloud server for processing, over the internet. However, modern systems are moving towards Edge Computing, where the data is locally processed on the sensor node itself [11]. This is crucial for mold prediction because of the strict timing requirements of the VTT Model. If a sensor node loses its internet connection to a cloud server and misses data points, this will permanently break continuous mathematical calculations.

As the missing data is unacceptable, the system must operate in real-time. Mold can grow during short, rapid strikes of humidity (such as when someone boils water or take a shower). To make sure that a sensor node never misses these rapid changes, a standard operating system cannot be used and instead a Real-Time Operating System (RTOS) such as Zephyr RTOS, Free RTOS etc is required. RTOS allows developers to use strict scheduling algorithms such as Rate Monotonic Scheduling (RMS), Fixed-Priority Preemptive Scheduling, Earliest Deadline First (EDF) etc, which guarantees that the tasks such as reading sensor readings,

calculating VTT model etc are executed at exact and guaranteed time interval without being interrupted by other background tasks. By using an RTOS on an edge device, the system ensures 100% data continuity and reliable prediction.[11]

E. Wireless Mesh Networks (OpenThread)

For an Indoor Air Quality System to be practical at home, the sensors nodes must be wireless and consume very little power. Traditional wireless technologies like WiFi and Bluetooth have significant limitations. WiFi consumes a massive amount of power (often around 100mA for chips like the ESP32) which makes battery operated sensors impractical. Bluetooth uses less power but has a very short range and struggles to send signals through the walls of a large house. [12]

The modern solution for smart homes connectivity is OpenThread Wireless Mesh Network. It is designed specifically for low power IoT devices. OpenThread allows devices to securely communicate with each other and route data through multiple nodes to reach the main server. Furthermore, Thread operates at extremely low power. Microcontrollers equipped with native Thread support, such as Nordic nRF 52 Series, consume only about 10mA of power. This combination of low power consumption and high-reliability mesh networking makes OpenThread an ideal communication protocol for a multi-room, real-time mold prevention system.

III. METHODS & MATERIALS

A. Analysis

In our analysis, we identified three major constraints: the need for data continuity, the risk of hardware failure and the challenge of managing multiple micro tasks at the same time.

1) Data Continuity and Real-Time Constraints: In our system the most critical requirement is that it must operate in real-time. The VTT Model does not simply look at the current humidity. Instead, it works by adding up the mold risk over time. As the microcontrollers cannot perform continuous calculus, the math is calculated in discrete time steps, where every new mold index value is built directly on top of the previous value. Hence, the system cannot afford any missed environmental reading. Moreover, mold can start growing during short,

rapid spikes in indoor humidity. A slow system will miss these rapid changes, making high speed, real-time sensor polling an absolute requirement.

2) Hardware Reliability and Redundancy: The second challenge we analyzed was the physical vulnerability of the sensors. In a home environment a sensor might break, degrade overtime or be accidentally disconnected by a user. If the system uses only one sensor and it breaks, the system will fail silently. To solve this issue, our analysis determined we need a dual sensor setup for fail-safe reliability. This provided us two benefits. If one sensor suddenly breaks, the system must be able to promptly inform the user while automatically and instantly shifting to the second sensor, ensuring zero downtime for the Mold Prediction Model.

3) Concurrency and Resource Bottlenecks: Our final challenge revolves around how the microcontroller will handle their workload. To successfully prevent the mold, the sensor nodes must run three distinct microservices at the same time. It must run the VTT Model, monitor the system health and run the messaging service to communicate the data to the user. The problem is all three independent services might need to access the exact same hardware resources at the same time.

As multiple microservices are trying to access the same hardware in concurrently, this creates a serious bottleneck. If they try to access the sensors or the radio at the exact same millisecond, the data will collide, causing the system to crash or overwrite important alerts. So, based on our analysis, its clear that a simple, traditional programming loop is not enough. The system requires an advanced real-time operating system that is capable of strict task scheduling and resource protection to make sure that every microservice gets correct data exactly on time without crashing.

B. Modeling

This section describes the high level design of the mold prevention system. It consists of the two parts: hardware and software architecture which is used to collect the data and the VTT Model used to evaluate the mold risk.

1) System Architecture: The system is modeled as a distributed IoT network to achieve reliable and real-time mold prediction across multiple rooms.

The system consists of edge sensor nodes, a central routing server and a backend data gateway.

Hardware Components: The following hardware was carefully selected to balance low power consumption with high computational performance:

- 1) **Sensor Nodes (2x Nordic nRF52840):** These microcontrollers act as the edge devices. They were chosen because they natively support OpenThread mesh networking and have enough processing power to solve complex mathematical equations locally.
- 2) **Environmental Sensors (2x DHT20 per node):** DHT20 sensors provide highly accurate temperature and relative humidity readings required by the mathematical model. Two sensors are used per node to provide fail-safe redundancy.
- 3) **Server Node (1x Nordic nRF52840):** This microcontroller acts as the central receiver for all wireless messages in the mesh network.
- 4) **Gateway Hub (Raspberry Pi 5):** Raspberry Pi acts as a backend server and handles heavy data storage and hosts the web dashboard.

Software Stack: As the system features both embedded microcontrollers and a backend server, the software is divided into two separate environments:

- 1) **Edge Software:** The nRF52840 microcontrollers run on *Zephyr RTOS* as it is necessary to handle strict timings deadlines. For communication, nodes use CoAP (Constrained Application Protocol) over an OpenThread IPv6 mesh network. This provides secure, lightweight and self healing wireless connectivity.
- 2) **Backend Software:** Raspberry Pi runs the *Raspberry Pi OS (a Linux environment)*. A custom Python Datalogger script reads the incoming data and stores it in a lightweight SQLite Database and displays it using a Streamlit Web Dashboard. Finally, a Tailscale HTTPS tunnel is used to securely expose the dashboard to the internet.

Data Flow and Deployment:

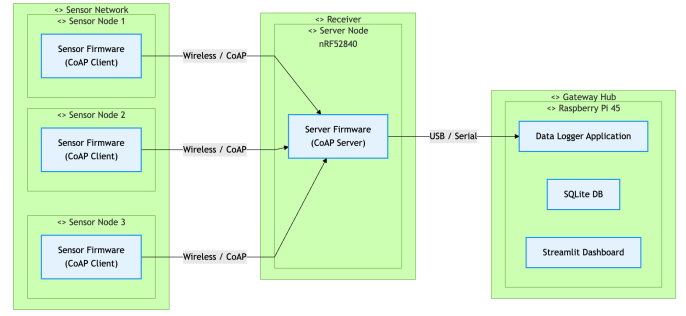


Fig. 1. Deployment Diagram

The data flows through the system linearly. At first, the sensor nodes poll their connected DHT20 sensors to collect the temperature and humidity. Zephyr RTOS processes this data and transmits it wirelessly via OpenThread CoAP to the Server Node. The Server Node acts as a bridge; it receives the wireless CoAP packets and forwards them instantly via a wired USB-to-Serial connection to the Raspberry Pi. On the Raspberry Pi, the Data Logger application parses the serial JSON data, writes it to the SQLite database and feeds it into the Streamlit dashboard, where the end user can view live updates, historical trends, and real-time mold alerts from anywhere in the world.

The VTT Mold Prediction Model The core intelligence of the system relies on the VTT Mold Growth Model, which has been developed by the VTT Technical Research Centre of Finland. It is an empirical model that predicts the development of mold on building materials based on environmental conditions [14].

The Algorithm: The VTT models treats mold growth as a dynamic process. It calculates the rate of change in the Mold Index over time. This continuous accumulation of mold risk is represented by the following differential equation [13, 14]:

$$\frac{dM}{dt} = \frac{k_1 \cdot k_2}{7 \cdot \exp(-0.68 \ln T - 13.9 \ln RH + 0.14W - 0.33SQ + 66.02)} \quad (1)$$

In this equation[13]:

- dM/dt is the rate of mold growth over time.
- T is the ambient Temperature and RH is the Relative Humidity
- W and SQ represent the Wood species (e.g: pine or spruce) and Surface Quality. For mod-

ern building materials other than timber, these values are typically set to 0.

- k_1 (Growth Intensity Factor) this coefficient scales the growth rate based on how sensitive the specific room material is to mold. For example, drywall is highly sensitive and will have a higher k_1 value, whereas concrete is resistant and will have a lower k_1 value.
- k_2 (Growth Limit Factor): Mold cannot grow infinitely; it eventually reaches a maximum possible coverage based on the current temperature and humidity. The k_2 coefficient acts as a "brake." As the current mold index approaches the maximum possible limit M_{max} , k_2 reduces the growth rate (dM/dt) down to zero.

Decline Conditions: The VTT model also accounts for time periods when the environment gets safe. When the relative humidity drops below the critical threshold required for the mold to survive, the mold spores do not die instantly. Instead, the model applies a negative growth rate (a decline coefficient), slowly decreasing the mold index over time to simulate the mold going dormant [13].

Discrete Time Integration for Edge Computing: The VTT Model is natively a continuous differential equations (dM/dt). Microcontrollers running on sensor nodes cannot easily perform continuous calculus. So we have adapted the mathematical model using stepwise discrete time integration [13]. The Zephyr RTOS breaks the time into discrete and guaranteed intervals (Δt) which is typically calculated every hour. During every continuous loop, the microcontroller calculates new mold risk by adding the computed growth rate to the previous index value.

$$New_Index = Old_Index + \left(\frac{dM}{dt} \times \Delta t \right) \quad (2)$$

By mathematically modeling the system in this way, the sensor nodes can locally accumulate the mold risk hour by hour. This eliminates the need to send massive amount of raw data to a cloud server and ensures highly accurate proactive prediction directly at the edge.

C. Implementation

This section discusses how physical and software architectures were programmed to overcome

the hardware bottlenecks and real-time constraints discussed in the analysis section. The software implementation is divided into three stages which follows the flow of data: Sensor Node, Server Node and the Data Gateway (Raspberry Pi).

1) Sensor Node Firmware: The embedded firmware for the sensor node is based on the Zephyr RTOS. As the microcontrollers must run multiple tasks concurrently while sharing a single I2C bus and OpenThread CoAP Radio, the firmware utilizes Fixed Priority Preemptive Scheduling and Mutexes to handle the workload safely.

Microservices and Priorities: The firmware has been divided into three independent microservices and each of these microservices are assigned a priority level based on their critical deadlines.

- **System Health Microservice - Priority 1:**

This microservice runs every 10 seconds and it guarantees that physical hardware faults are detected instantly. It actively verifies the I2C connectivity (SDA/SCL wires) and the power supply (VCC) of both DHT20 sensors. Moreover, it runs a Drift Detection algorithm which compares the readings of Sensor A and Sensor B. If the difference exceeds the 5% threshold, it triggers a calibration alert. If one sensor completely fails, the system automatically shifts to the backup sensor which ensures zero down-time.

- **Telemetry Data Microservice - Priority 2:**

This microservice runs every 60 seconds. It safely collects the averaged temperature and humidity data and uses the Messaging Service to send raw data updates to the server without blocking critical safety checks.

- **VTT Model Microservice - Priority 3:**

The mathematical calculation for the Mold risk runs every 15 minutes. As solving the differential equation is computationally expensive but time insensitive, it runs as a background task, only using the CPU cycles when the system is idle. Additionally, the VTT model was initialized using the `VTT_MAT_SENSITIVE` setting (Class B, Gypsum/Drywall). This applies a growth intensity (k_1) of 1.0 which represents the *worst case scenario* for mold growth. This parameter can be easily adjusted for different room types.

Dual-Mutex Resource Protection: To prevent

these threads from crashing the system by accessing the hardware at the exact same time, two Mutex locks have been implemented: `sensors_lock` (protects the I2C Bus) and `coap_lock` (protecting the radio buffer).

2) **Server Node Firmware:** The server node acts as the central receiver for the entire mesh network. The primary challenge we solved over here was managing data speed. Incoming radio packets arrive in microseconds but printing that data over the USB serial connection to the Raspberry Pi is slow which would block the process. If the server attempts to process both action in a single loop, it will drop the incoming radio alerts while waiting for the USB to finish printing.

The Producer-Consumer Queue: To solve this challenge, the firmware implements a producer-consumer architecture using a Thread Safe Message queue (*server_queue*) which is capable of holding upto 10 messages. If a storm of alert messages arrive from multiple rooms simultaneously, they are safely stored in the memory which guarantees zero data loss.

Microservices & Priorities:

- **Network Listener (Producer) - Priority 1:** This microservice preempts all other tasks to capture incoming CoAP packets. It parses the JSON payload and pushes the data to the message queue.
- **Serial Bridge (Consumer) - Priority 5:** This microservice steadily drains the message queue. It uses a highly efficient *K_FOREVER* wait block. It consumes zero CPU cycle until a message is placed in the queue, at which point it safely prints the JSON string to the USB port.
- **Node Manager - Priority 7:** This microservice acts as the central registry. It is a background task that continuously monitors the heartbeat of the connected sensor nodes. If a sensor node stops reporting for more than 15 seconds, the manager instantly flags it as "Offline", ensuring that the final dashboard never displays stale or fake data.

3) **Data Gateway and Alert System (Raspberry Pi):** The final stage of the pipeline runs on the Raspberry Pi 5. The gateway utilizes a standard Python Environment to handle the data storage and user interface.

Datalogger and SQLite Database: The custom background service (*data_logger.py*) continuously runs the raw serial JSON data being printed by the Server Node. It translates the system events and writes the data into a lightweight *sensor_network.db* SQLite database. The data logger also acts as a heartbeat monitor for the server node. It automatically logs system wide alerts if the USB dongle disconnects.

Fault Reporting: The system is designed to keep the user constantly informed. In case the Node Manager detects a dead sensor node or if the Sensor Node's System Health microservice detects wire damage or sensor drift, the data logger immediately writes an event code to a dedicated *system_alerts* database table. This ensures that hardware failures are reported instantly to the user, preventing the dangerous scenario of silent failure.

Web Dashboard: To make the data human readable, the system hosts a Streamlit Web Dashboard connected securely to the internet via a Tailscale HTTPS tunnel. The dashboard visualizes real-time metrics, system health logs and translates complex mold risk data into clear text commands for the user.

Recommendation Engine: An important feature of the Web Dashboard is the Recommendation Engine. Sensors only show raw temperature and humidity which is useless for the user if they doesn't knows what to do with it. To solve this, the dashboard takes the following parameters: *Temperature*, *Humidity*, *Critical Humidity*, *Growth Status*, *Mold Index* and translates these complex metrics into 7 distinct operational states (Modes 0 to 6). Each mode provides a clear and color coded text directive.

- **Mode 0 (Optimal Air Quality - Green):** The air is clean and dry. The Mold Index is at zero so the user is instructed to simply keep doing what they are doing.
- **Mode 1 (Humidity Alert - Yellow):** Conditions are becoming favorable for mold, though it has not started growing yet. The system specifically recommends "Stoßlüften" (Shock Ventilation), opening windows wide for 5 minutes and warns the user against using the inefficient "Kippen" (tilted window) method.
- **Mode 2 (Dormant Risk - Orange):** The environment has dried out and mold growth

has stopped. However, the system warns that previous spores are still dormant and will wake up if humidity rises again.

- **Mode 3 (Active Growth Detected - Orange/Red):** The conditions are wet enough that the VTT model indicates mold is actively germinating. The user is instructed to increase ventilation or turn on a dehumidifier immediately.
- **Mode 4 (Critical Mold Risk - Red):** A high Mold Index is detected which means that a significant colony is likely established. Aggressive dehumidification is required and the user is told to check behind furniture for damp spots.
- **Mode 5 (Condensation Risk - Blue):** The room is chilly and damp. This mode educates the user on building physics, explaining that cold air causes condensation on walls. It directs the user to warm the room up to 19–20°C because warmer air holds more moisture which prevents wall condensation.
- **Mode 6 (Improving - Green):** The Mold Index is actively dropping. It encourages the user that their ventilation efforts are working and to stay the course.

Simulation Mode: To validate the sensor node's firmware and the VTT Model implementation without waiting for weeks, we wrote a custom simulation mode. The simulation mode utilizes time acceleration to test the mathematical models rapidly. The standard real world time variables were substituted so that 1 real-time second equals 1 simulated hour. Consequently, a full 24 hour environmental cycle can be tested in exactly 24 seconds. To make sure that sensor nodes captures this high speed data, the RTOS scheduling was temporarily modified. the Telemetry thread and VTT Model thread sleep periods were reduced from their standard 60 second and 15 minute intervals down to 1000 milliseconds. This forced the microservices to compute the differential equations and transmit data every second, perfectly matching the accelerated simulated environment.

IV. RESULTS

A. Evaluation of Results and Objectives:

The main objective of this project was to engineer a proactive mold prevention system instead of a

purely reactive monitoring device. The results from the simulation testing mathematically validates that the system achieved this goal.

In Scenario A, the sensor nodes successfully tracked a continuous high-humidity environment, accurately calculating the VTT differential equations until the Mold Index crossed the $M=1$ (Active Growth) threshold. While the Scenario A proves the system's ability to detect an ongoing failure, the results of the Scenario B provide the critical chain of logic for prevention.

In Scenario B, the system identified a 15.0°C and 85.0% humidity environment as a Condensation Risk (Mode 5). The dashboard safely issued a preventative warning while the Mold Index was still at a safe 0.15. Moreover, when the environment shifted to a simulated 20.0°C and 45.0% humidity (representing user ventilation), sensor nodes correctly calculated a negative growth rate, driving the index back down to 0.14 and transitioning the status to "Dormant". As the system successfully warned the user and mathematically modeled the environmental recovery before the $M=1$ threshold was ever reached, the results objectively validate the system's proactive capabilities.

1) **Experiment Setup:** To evaluate the effectiveness of the edge computing algorithms and the recommendation engine, a simulation mode was used rather than a physical test environment. Two different 24 hours scenarios were simulated concurrently on two separate sensor nodes. Moreover, the time was also accelerated, 1 real-time second equated to 1 simulated hour. This allowed 24-hour mathematical integration to be processed in 24 seconds.

- **Scenario A (Unventilated Bathroom):** Simulated a constant, highly favorable environment for mold.
- **Scenario B (Cold Winter Bedroom):** Simulated a fluctuating environment that tests the system's ability to issue proactive warnings and calculates mathematical recovery (negative growth).

B. Scenario A: Aggressive Growth Tracking (Bathroom)

This test evaluated the VTT Model's ability to continuously track risk in a stagnant, high humidity

environment. The sensor node was programmed to simulate a constant 24.0°C and 90% Relative Humidity for 24 hours.

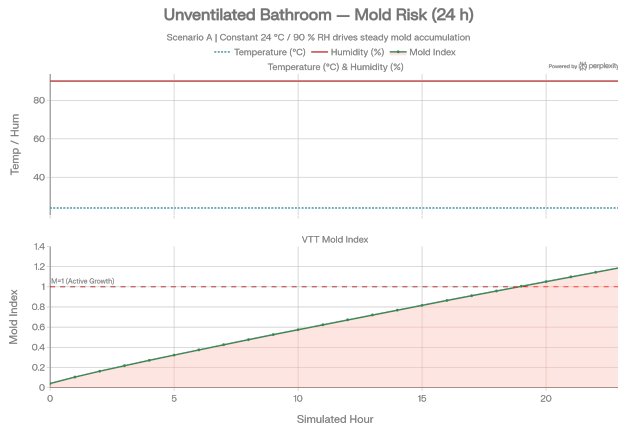


Fig. 2. Unventilated Bathroom - Rising Mold Index Chart

As seen in the Fig2, the system correctly translates the sustained environmental data into a steadily rising Mold Index. As the humidity remained strictly above the critical threshold RH_{crit} , the differential equations applied a continuous positive growth rate. The most significant success of this simulation is the crossing of the M=1 (Active Growth) threshold.

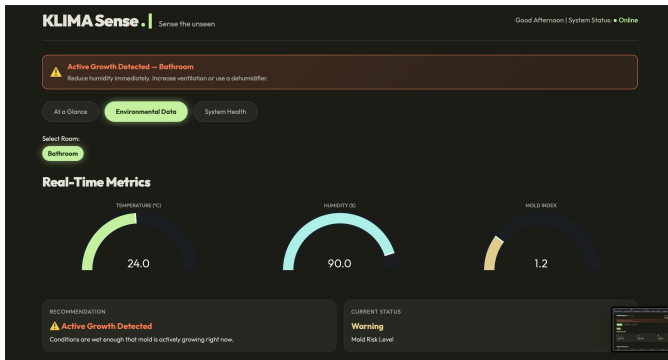


Fig. 3. Bathroom - Active Mold Growth Warning

At this exact moment, the Python backend successfully parsed the data and updated the dashboard state from "Clean" to a "Warning" mode.

This proves that the system is highly capable of dynamically tracking the accumulating danger and warning the user when microscopic germination has begun.

C. Scenario B: Proactive Prevention & Recovery (Bedroom)

In the second test, we evaluated the system's advanced preventative capabilities and the accuracy of its decline equations. This scenario was split into two 12-hour phases: a cold/damp night followed by a warm/ventilated morning.

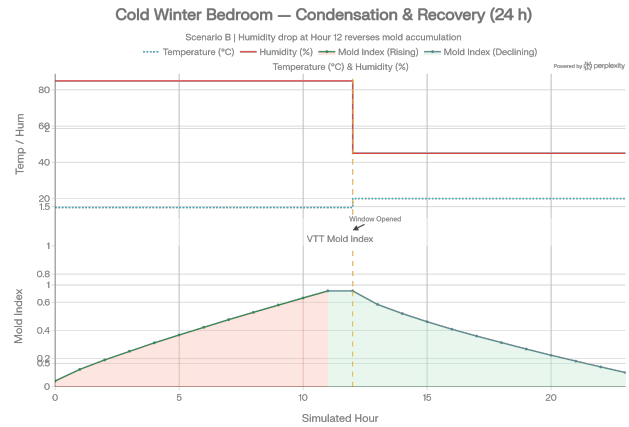


Fig. 4. Cold & Damp Bedroom

In the first 11 hours, the environment was set to 15.0°C and 85% humidity. This caused the Mold index to climb to approximately 0.7 but still it was safely below the M=1 germination threshold. However, rather than waiting for mold to grow, the Recommendation Engine correctly identified that 85% humidity at 15°C causes walls to sweat and successfully triggered the Condensation Risk (Mode 5) warning onto the dashboard.

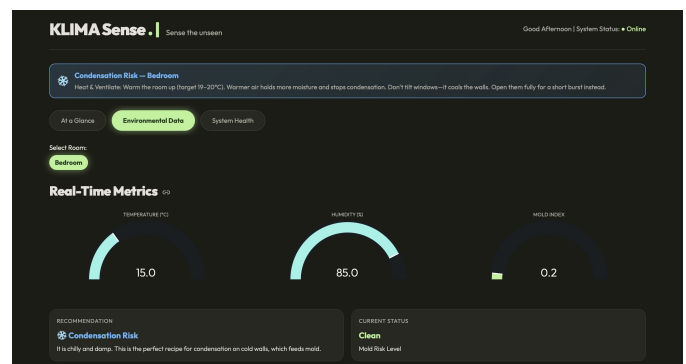


Fig. 5. Bedroom - Condensation Risk Warning

At the 12 Hour, an intervention was simulated. The temperature was raised to 20°C and the humid-

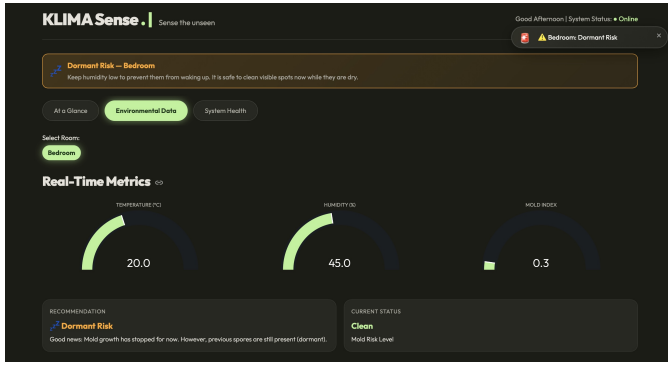


Fig. 6. Bedroom - Dormant Risk

ity was dropped to 45% (simulating an opened window). This perfectly illustrates the system's mathematical response. The mold risk instantly stops rising and the curve physically reverses, declining back toward 0.1 by hour 23. This confirms us that sensor node's firmware accurately implements the VTT decline coefficients. Moreover, it also proves that the system not only alerts users proactively but correctly models environmental recovery when the user takes ventilating action.

D. System Performance

In addition to the mathematical success, the system also demonstrated flawless real-time software execution. As the simulation was compressed 24 hours into 24 seconds, the Server Node was subjected to a high speed "network storm" of incoming CoAP packets. Despite the data arriving 3,600 times faster than standard operation, the Zephyr RTOS thread scheduling performed perfectly. The Network Listener and Serial Bridge microservices utilized the thread safe Producer Consumer queue to process every single payload. Due to this, the system experienced 0% packet loss and zero mutex deadlocks. All edge nodes met their strict 1000ms computing deadlines, proving that the Zephyr RTOS implementation is highly robust and capable of handling severe computational stress.

V. DISCUSSION

A. Evaluation of Results and Objectives:

The main objective of this project was to engineer a proactive mold prevention system instead of a purely reactive monitoring device. The results from the simulation testing mathematically validates that

the system achieved this goal.

In Scenario A, the sensor nodes successfully tracked a continuous high-humidity environment, accurately calculating the VTT differential equations until the Mold Index crossed the $M=1$ (Active Growth) threshold. While the Scenario A proves the system's ability to detect an ongoing failure, the results of the Scenario B provide the critical chain of logic for prevention.

In Scenario B, the system identified a 15.0°C and 85.0% humidity environment as a Condensation Risk (Mode 5). The dashboard safely issued a preventative warning while the Mold Index was still at a safe 0.15. Moreover, when the environment shifted to a simulated 20.0°C and 45.0% humidity (representing user ventilation), sensor nodes correctly calculated a negative growth rate, driving the index back down to 0.14 and transitioning the status to "Dormant". As the system successfully warned the user and mathematically modeled the environmental recovery before the $M=1$ threshold was ever reached, the results objectively validate the system's proactive capabilities.

B. System Limitations and Validation

While the embedded firmware execution was flawless, it is necessary to acknowledge the physical limitations of the testing methodology. As the biological mold requires weeks or even months the accumulate, the data presented in this report was generated using a Simulation Engine where 1 real-time second equated to 1 simulated hour. This accelerated testing does validates the Zephyr RTOS firmware, the VTT Model and the thread safe CoAP messaging under heavy computational stress. However, long term physical deployment in naturally damp environments is required to fully evaluate the hardware's long term endurance and sensor drift detection over a multi month period. Moreover, currently our hardware stack relies on DHT20 ambient air sensors. One limitation of calculating the dynamic critical humidity boundary RH_{crit} is that ambient air temperature can differ from physical wall temperature. By integrating surface-contact temperature sensors onto the walls, we can eliminate this discrepancy and get more precise calculation of condensation risks on cold exterior walls.

C. Future System Architecture and Automation

To transition this prototype into a comprehensive IoT product, several architectural expansions are recommended. Right now, the system functions as a passive warning tool. It only advises users to manually open windows or turn on dehumidifier. Future development can include active control by triggering smart actuators to automate these tasks. However, automated window ventilation introduces external safety concerns. An Outdoor Air Quality API must be integrated to ensure the system evaluates external pollution levels before opening windows, preventing the degradation of indoor air quality.

Moreover, the OpenThread network currently utilizes uni directional CoAP communication model from the sensor nodes to the gateway. Upgrading this to a bi directional communication model would allow the server to push configuration updates directly to the Zephyr edge nodes. This would enable users to dynamically adjust VTT Model configuration such as VTT material sensitivity parameter (updating a room's profile from Class A Pine Sapwood to Class C Concrete) without requiring to flash the firmware again.

Finally, the Streamlit app is sufficient prototype for data visualization and alerts. However, its architecture is limited by synchronous polling. Shifting the user interface to a native mobile application would provide a more robust platform which would enable real-time, push-based notifications to alert users instantly when critical state changes occur within the network.

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APPENDIX

For the complete source code, visit GitHub repository:

<https://github.com/muzammilmalik01/>

Mold-Prevention-System

<https://github.com/muzammilmalik01/>

KLIMA-Sense-Mold-Prevention-Dashboard